BLOWING UP THE R-TH FITTING IDEAL MAKES A MODULE FINITE LOCALLY FREE OF RANK R ON A SCHEME THEORETICALLY DENSE OPEN

THE STACKS PROJECT AUTHORS

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Editorial note. Statement span 9339-9356 and proof span 9358-9397 come from the halfA pool card operations/pool-cards/stacks-divisors-halfA.json and were re-read in the pinned chapter file before this card was written.

Editorial note. The proof contains the author's parenthesis noting that the complement of the Fitting subscheme may be bigger than the original open, which is why the density step is needed; it is preserved verbatim.

Editorial note. 6 cross-references occur inside the counted spans; they are listed in external_references as navigation pointers only.

Allowed standard prerequisites (editorial).

- The r-th Fitting ideal of a finitely presented module is of finite type, and its restriction to a locus where the module is finite locally free of rank r is the whole structure sheaf, while the (r-1)-st Fitting ideal restricts to zero there. Cited at proof lines 9358-9364 and 9369-9372. Gives admissibility of the blowing up and shows that the smaller Fitting ideal vanishes identically because the open is scheme theoretically dense.
- The definition of an admissible blowing up, and the statement that a module becomes finite locally free of the expected rank after blowing up the appropriate Fitting ideal. Cited at proof lines 9364-9366 and 9378-9382. Supplies the notion of admissibility and the conclusion about the strict transform.
- The pullback of a finitely presented module is finitely presented, cited from the modules chapter. Cited at proof lines 9390-9392. Shows that the kernel of the surjection onto the strict transform is finitely presented once the pullback is split.
- The vanishing of the higher Tor of a module whose Fitting ideals behave as here, cited from the more-algebra chapter. Cited at proof lines 9395-9397. Gives the statement about perfectness and tor dimension.

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Source reference. Original definition 080K: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10185–10185. Same-chapter reference used in the retained author text at source line(s) 9364. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 0C3G: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 15191–15191. Same-chapter reference used in the retained author text at source line(s) 1441, 9362, 9367, 9372. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 0C3E: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 15189–15189. Same-chapter reference used in the retained author text at source line(s) 1450, 9360. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 0CZQ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 16285–16285. Same-chapter reference used in the retained author text at source line(s) 9378. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 01BQ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1735–1735. Same-chapter reference used in the retained author text at source line(s) 9391. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 0F7M: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 19012–19012. Same-chapter reference used in the retained author text at source line(s) 9396. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Lemma 0.1. *Let X be a scheme. Let $\mathcal{F}$ be a finitely presented $\mathcal{O}_X$-module. Let $U \subset X$ be a scheme theoretically dense open such that $\mathcal{F}|_U$ is finite locally free of constant rank r. Then*

1. *the blowup $b : X' \to X$ of X in the rth Fitting ideal of $\mathcal{F}$ is U-admissible,*
2. *the strict transform $\mathcal{F}'$ of $\mathcal{F}$ with respect to b is locally free of rank r,*
3. *the kernel $\mathcal{K}$ of the surjection $b^*\mathcal{F} \to \mathcal{F}'$ is finitely presented and $\mathcal{K}|_U = 0$,*
4. *$b^*\mathcal{F}$ and $\mathcal{K}$ are perfect $\mathcal{O}_{X'}$-modules of tor dimension ≤ 1.*

Proof. The ideal $\mathrm{Fit}_r(\mathcal{F})$ is of finite type by Lemma 0C3E and its restriction to U is equal to $\mathcal{O}_U$ by Lemma 0C3G. Hence $b : X' \to X$ is U-admissible, see Definition 080K.

By Lemma 0C3G the restriction of $\mathrm{Fit}_{r-1}(\mathcal{F})$ to U is zero, and since U is scheme theoretically dense we conclude that $\mathrm{Fit}_{r-1}(\mathcal{F}) = 0$ on all of X. Thus it follows from Lemma 0C3G that $\mathcal{F}$ is locally free of rank r on the complement of subscheme

cut out by the rth Fitting ideal of $\mathcal{F}$ (this complement may be bigger than U which is why we had to do this step in the argument). Hence by Lemma 0CZQ the strict transform

$$b^*\mathcal{F} \longrightarrow \mathcal{F}'$$

is locally free of rank r. The kernel $\mathcal{K}$ of this map is supported on the exceptional divisor of the blowup b and hence $\mathcal{K}|_U = 0$. Finally, since $\mathcal{F}'$ is finite locally free and since the displayed arrow is surjective, we can locally on X' write $b^*\mathcal{F}$ as the direct sum of $\mathcal{K}$ and $\mathcal{F}'$. Since $b^*\mathcal{F}'$ is finitely presented (Modules, Lemma 01BQ) the same is true for $\mathcal{K}$.

The statement on tor dimension follows from More on Algebra, Lemma 0F7M. □